

Communication and Collaboration in Environmental Science Group Assignment: Interactive Environmental Data Dashboard
Spring term 2026
laup, V2, 16.03.2026

Group size: 2–3 students

Objective: Develop an interactive dashboard that explores environmental data with spatial components and communicates insights through effective visual storytelling.

Project Overview: You will design and implement an interactive dashboard using open environmental data. The dashboard must include a map-based view and linked visualizations, and be extended with one advanced feature.
The project will be structured in two phases

Phase 1 — Core Dashboard: With coaching support and practical guidance, build a basic dashboard that includes:

- A map view displaying spatial data
- Several linked plots (e.g., charts that respond to map selections)
- Clear interaction between components

Data requirements:

- Use datasets from <https://opendata.swiss/> (or comparable open sources)
- Must have an environmental context
- Must include a spatial dimension
- May include a temporal dimension

Phase 2 — Dashboard Extension: Extend your dashboard by implementing one of the following topics. The extension should meaningfully enhance user interaction or insight.

- Animation
- Participative mapping
- User-generated content (UGC)
- Uncertainty visualization
- Another mapping technique (Chapters 15–22)

Mid-Semester Coaching Session (Work in Progress)

You will present your current progress to peers and instructors.

Prepare:

- A short demo of your dashboard (current state of work, does not have to be finished but should illustrate initial ideas)
- Show at least the datasets you are using (properties, resolutions, data quality issues you may have)
- 2–3 questions for feedback or clarification

Final deliverables

1. Technical Project Report (Computational Notebook)

- Length: 1500-2500 words of text (excluding figures and code snippets), in a rendered form and as a formatted pdf. Provide an URL to a git-Repo (e.g. GitHub)
- The report must include:
 - Dashboard context and goals (including chosen extension topic)
 - Description and sources of datasets used
 - Storytelling concept and intended insights
 - Key dashboard functionalities
 - Executable code for the dashboard
 - A declaration of individual students' contributions made within the project
 - A declaration of AI use

2. Screencast/Video (3–4 minutes)

Create a short video that shows your dashboard in action:

- Demonstrates key dashboard features
- Explains functionality via voice-over

Deadlines

22.04.2026: Presentation work in Progress

31.05.2026: Submission of final project report, plus supplements

Evaluation Criteria

- Data suitability and preparation
- Dashboard design and interactivity
- Implementation of extension topic
- Clarity of storytelling and insights
- Technical quality and reproducibility
- Quality of report and screencast